

Code Conventions for the Java Programming Language

These Conventions reflects the Java language coding standards presented in the [Java Language Specification](#), from Sun Microsystems.

There are many conventions which are provided in this specifications will see some important once.

• File Names

File Type	Suffix
Java source	.java
Java bytecode	.class

• Naming Conventions

Identifier Type	Rules for Naming	Examples
Packages	The prefix of a unique package name is always written in all-lowercase ASCII letters and should be one of the top-level domain names, currently com, edu, gov, mil, net, org, or one of the English two-letter codes identifying countries as specified in ISO Standard 3166, 1981. Subsequent components of the package name vary according to an organization's own internal naming conventions. Such conventions might specify that certain directory name components be division, department, project, machine, or login names.	com.sun.eng com.apple.quicktime.v2 edu.cmu.cs.bovik.cheese
Classes	Class names should be nouns, in mixed case with the first letter of each internal word capitalized. Try to keep your class names simple and descriptive. Use whole words-avoid acronyms and abbreviations (unless the abbreviation is much more widely used than the long form, such as URL or HTML).	class Raster; class ImageSprite;
Interfaces	Interface names should be capitalized like class names.	interface RasterDelegate; interface Storing;
Methods	Methods should be verbs, in mixed case with the first letter lowercase, with the first letter of each internal word capitalized.	run(); runFast(); getBackground();
Variables	Except for variables, all instance, class, and class constants are in mixed case with a lowercase first letter. Internal words start with capital letters. Variable names should not start with underscore _ or dollar sign \$ characters, even though both are allowed. Variable names should be short yet meaningful. The choice of a variable name should be mnemonic- that is, designed to indicate to the casual observer the intent of its use. One-character variable names should be avoided except for temporary "throwaway" variables. Common names for temporary variables are <i>i</i>, <i>j</i>, <i>k</i>, <i>m</i>, and <i>n</i> for integers; <i>c</i>, <i>d</i>, and <i>e</i> for characters.	<pre>int i; char c; float myWidth;</pre>
Constants	The names of variables declared class constants and of ANSI constants should be all uppercase with words separated by underscores ("_"). (ANSI constants should be avoided, for ease of debugging.)	<pre>static final int MIN_WIDTH = 4; static final int MAX_WIDTH = 999; static final int GET_THE_CPU = 1;</pre>

• Some Programming Practice

1. Try to keep the names of classes,interface,variables and constants to be meaningful and related.
2. You can name i,j,k,a ...etc as variables for temporary variable.
3. Files longer than 2000 lines are cumbersome and should be avoided.
4. The first non-comment line of most Java source files is a package statement. After that, import statements.
5. Try to write comments like documentation comments,others.

6. One declaration per line is recommended since it encourages commenting. In other words,

```
int level; // indentation level
int size;  // size of table
```

is preferred over

```
int level, size;
```

7. Try to initialize local variables where they're declared. The only reason not to initialize a variable where it's declared is if the initial value depends on some computation occurring first.